



Algebra by the Eighth Grade: The Association Between Early Study of Algebra I and Students' Academic Success

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Abstract

Using the High School Longitudinal Study of 2009, we examine the association between taking Algebra I by the eighth grade and students' later achievements and advanced mathematics course-taking. To explore our research questions, we employ three distinct methods—regression analysis, fixed effects models, and propensity score matching. In all our analytic models, we consistently found that students who have taken Algebra I by the eighth grade are more likely to have higher achievements in the ninth and eleventh grades and to take classes beyond Calculus during high school. Although the study is correlational, considering the increasing number of students taking Algebra I by the eighth grade, its findings have meaningful implications for education policy, research, and practice.

Keywords Algebra · Achievement · Course-taking · Calculus

Introduction

School districts have focused on strengthening and increasing mathematical literacy because a solid foundation in math is an important component in student success (Strauss, 2012). Among the many mathematics curricula reforms and initiatives, considerable emphasis has been placed on Algebra I (hereafter algebra), a gatekeeper

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course, which is viewed as a course that leads to higher achievement scores (Filer & Chang, 2008) and postsecondary education success (Dougherty, Goodman, Hill, Litke, & Page, 2017). Not surprisingly, various efforts such as universal algebra and mathematics acceleration policies have been introduced to offer early exposure to algebra (Nomi & Raudenbush, 2016). As a result, the number of students enrolled in algebra has doubled in the past two decades (Loveless, 2013).

Recognizing that mathematics has a hierarchical structure in which some strata depend on students completing earlier courses, the issue of when the algebra course is taken may be critical (Loveless, 2008). Proponents of early algebra argue that by completing algebra before high school, students can take Geometry in their first year of high school, Algebra II as sophomores, Precalculus as juniors, and Calculus or other advanced math courses in their senior year (Loveless, 2008). This linear-structured course sequence has created some positional advantages for students who start their math trajectories earlier than their peers (Schneider, Swanson, & Riegle-Crumb, 1998). While some students who start algebra in high school can advance to Calculus or other advanced math courses—students can take either two math courses concurrently or additional math courses online—the opportunity is limited (Spencer, 2016) and doing so is challenging (Loewus, 2014). Intuitively, the idea that early algebra should precede advanced math courses makes sense, but relatively little attention has been given to the association between early algebra and advanced math courses, and most of the studies have been largely descriptive (Paul, 2005; Spielhagen, 2006). Moreover, although studies using national representative datasets—the National Educational Longitudinal Study (NELS) of 1988 (Rickles, 2013; Smith, 1996) or the Longitudinal Study of American Youth (LSAY) of 1987 (Ma, 2000)—provide valuable evidence about the role of algebra in student achievement, they used data compiled three decades ago, and most did not sufficiently account for selection bias (Stein, Kaufman, Sherman, & Hillen, 2011). This hinders our ability to build a plausible and coherent policy around algebra. In fact, considering the different policy, curriculum emphases, and school resources compared with that in early 1990s, evidence from the existing literature is insufficient to support current educational practices and policies. For this reason, the number of students who take algebra before high school varies considerably between states and even within school districts (Walston & McCarroll, 2010).

This paper fills the gap and contributes to the literature by examining the association between early algebra and student achievement and course-taking, using three distinct analyses of the most recent national dataset. By utilizing the most recent national representative dataset of the High School Longitudinal Study of 2009 (HSLS:09), this study investigates the relationship between early algebra (by eighth grade) and student achievement and advanced mathematics course-taking. Furthermore, this study conducted three distinct analyses, naïve to more rigorous—ordinary least square (OLS) and probit regression, fixed-effects models, and propensity score matching—to better understand and to efficiently isolate the association between early algebra and student educational success. The study's overarching goal is to consider the following research questions:

- 1 What is the association between early algebra and student achievement (ninth- and eleventh-grade math scores)?
- 2 What is the association between early algebra and students' advanced mathematics course-taking (whether students have taken Calculus or above during high school)?

Early Algebra and Student Achievement

Proponents of early algebra say that offering algebra to students in eighth grade or earlier improves math achievement scores (Gamoran & Hannigan, 2000; Ma, 2000). By using NELS:88, Gamoran and Hannigan (2000) found that students who took eighth-grade algebra achieved higher mathematics scores in both eighth and tenth grades compared with their peers who started algebra later. By analyzing the same dataset, Rickles (2013) demonstrated that students who completed algebra by eighth grade tend to score higher on their twelfth-grade math tests. Based on LSAY:87, Ma (2000) similarly found that students who take algebra in their seventh or eighth grade have higher math test scores than those who take it later in high school. A study by Kurlaender, Reardon, and Jackson (2008), using data from three California school districts, descriptively found that students who complete algebra by eighth grade have a higher graduation rate compared with other students in high school. Thus, recognizing the importance of algebra, states such as California implemented policies that set algebra as mandatory for all eighth graders (Spielhagen, 2006). From 1997 to 2012, all students in California were required to take algebra and were assessed through state-level testing. This increased the percentage of eighth graders taking algebra in California from 32% in 2003 to 59% in 2011 (Domina, McEachin, Penner, & Penner, 2015).

Even so, analysis of the results of universal algebra policies or mathematics acceleration programs was less than encouraging (Liang, Heckman, & Abedi, 2012). As shown by Liang et al. (2012), in California, where eighth-grade algebra was mandatory, approximately 60% failed to score “proficient” in the Algebra California Standards Test. As a result, in 2013, California backed away from the mandatory algebra requirement (Dougherty et al., 2017). A study in North Carolina, where some school districts had initiated similar algebra acceleration policies, found a negative impact of early algebra instruction. Students who were accelerated into algebra courses in eighth grade, compared with those who had no early algebra exposure, tended to have lower scores not only in algebra tests but also in other follow-up mathematics courses in later years (Clotfelter, Ladd, & Vigdor, 2015). Thus, despite the policies’ best intentions, mandating that all students take eighth-grade algebra, regardless of their readiness, has only a small positive—or small negative—effect on student achievement.

Early Algebra and Advanced Course-Taking

Prior research shows that taking more advanced math courses—courses beyond Algebra II—can positively impact educational outcomes such as standardized test scores (Attewell & Domina, 2008), high school completion (Long, Conger, & Iatarola, 2012), and postsecondary performance (Adelman, 2006). A study from Georgia revealed that students who completed more credits in advanced math courses during their secondary years spent less time on lower-level postsecondary courses and graduate from college at a higher rate compared with other students (Ackerman, Kanfer, & Calderwood, 2013). Sadler and Tai (2007) demonstrated that students who have taken Calculus in high school, compared with students who stop at precalculus or below, achieve higher grades in college-level mathematics courses than other students. More importantly,

the students had cross-subject benefits such as higher grades in college-level science courses. It is not surprising, then, to see a dramatic increase in students taking high school advanced math courses. For instance, between 1990 and 2009, the percentage of high school graduates who had taken Calculus increased from 7 to 16% (NCES, 2016).

Although proponents of early algebra argue that increasing the number of students in early algebra will lead to more students taking advanced math courses and be better prepare them for college (Loveless, 2008; Spielhagen, 2006), few empirical studies have closely examined these claims (Ma, 2000; Paul, 2005; Smith, 1996; Spielhagen, 2006). A study by Spielhagen (2006), based on data from a large school district, found that students who take eighth-grade algebra stay longer in the math course pipeline and take more advanced courses in high school than other students. Using data from five urban high schools, Paul (2005) consistently found that students enrolled in algebra in eighth grade were more likely to enroll in advanced courses, such as biology, chemistry, physics, and Algebra II, compared with students who took algebra later. However, a study from North Carolina revealed that students do not benefit equally from being accelerated to algebra (Clotfelter et al., 2015). Indeed, moderately and high-performing students were found to have a better chance of enrolling in Calculus after eighth-grade algebra, while relatively low-performing students, even with early exposure to algebra, were less likely to progress to high school Calculus.

Though the literature provides some evidence on the role of early algebra on students' educational pathways, to date, most of the studies have relied on descriptive or *t* test statistics to gauge the percentage of students enrolled in advanced courses after completing early algebra. To close the gap in the literature and to provide strong empirical evidence for the role of early algebra has on educational pathways, this study examines the association between early algebra, student achievement, and course-taking patterns, using the most recent national representative dataset. Furthermore, by using three distinct methods, the study attempts to address validity threats due to selection bias that undermined many of the earlier studies (Rickles, 2013).

Tracking in Mathematics

To have more students take algebra before high school, middle school has been utilizing tracking practice, also known as ability grouping, to group students into different courses, programs, or schools (Lessard, Larose, & Duchesne, 2020). In particular, many US middle schools use a two-tiered tracking system in mathematics; most students will take a grade-level mathematics class, while other students attaining a higher achievement level in math will take algebra in eighth grade. This practice has been used widely across the globe in places such as the USA, the UK, Belgium, and Australia (Hall, 2014; Houtte, Demanet, & Stevens, 2012). While countries like Germany utilize the results of tracking to place students in different schools based on their academic achievement and needs (Schofield, 2010), tracking in most countries, including the USA, refers to the practice of grouping students into different levels and streams of courses (Ansalone, 2010).

Tracking or grouping students based on their academic background has been in the education system since 1867 (Ansalone & Biafora, 2004); nonetheless, it is one of the most

controversial practices in education (Ansalone & Biafora, 2010). While supporters of tracking argue that students learn better when they are surrounded by peers with similar academic backgrounds (Ansalone, 2010), opponents of tracking believe that tracking increases educational inequality among students (Pallas, Entwisle, Alexander, & Stluka, 1994). In fact, scholars have found that tracking practices and assignments to track students are regularly biased by racial, ethnic, and socioeconomic status (Maaz, Trautwein, Lüdtke, & Baumert, 2008). To address such concerns, a few districts have declared “detracking” practices and set high expectations for all students, but the outcomes of this policy remain unclear (Domina, Hanselman, Hwang, & McEachin, 2016).

Despite such concerns regarding equity and equality, many schools in the USA are still using tracking to group students into different tracks and courses that may lead to different educational outcomes (McEachin, Domina, & Penner, 2020). Moreover, there is a lack of consensus in the literature on the effects of tracking on students’ education and whether or not it widens social inequalities (Schiller, Schmidt, Muller, & Houang, 2010). Scholars have also found that most teachers and parents tend to support such tracking (Ansalone & Biafora, 2004, 2010); it is for these reasons that most school leaders are reluctant to make any changes that might make their community members unhappy (Biafora & Ansalone, 2008). When considering that tracking is a common practice in mathematics, the findings of the present study have wider implications for mathematics education across the globe.

Method

Data and Sample

This study utilized a restricted version of the High School Longitudinal Study of 2009, which is an ongoing study from the National Center for Education Statistics (NCES) that tracks a nationally representative sample of secondary students. The HSLs:09 collected information on ninth-grade students from both public and private schools in the USA beginning in fall 2009 and using a stratified, two-stage random sampling design where schools were the primary sampling units. The data consists of approximately 21,440 ninth-grade students nested within 944 participating high schools in the fall of 2009, an average of about 27 students per school. The first follow-up of HSLs:09 was conducted in spring 2012 when students were in the eleventh grade, and the second follow-up was held in spring 2016. Surveys of students, parents, math and science teachers, school administrators, and counselors are included in the dataset.

Our sample includes all students surveyed in both the base year (2009) and the second follow-up (2011); we exclude students who are missing the following dependent variables: ninth-grade math score, eleventh-grade math score, and transcript information regarding their math course-taking. Students who also had not taken algebra during their secondary schooling were excluded, which left us with about 15,390 students. Missing data percentages for all analytic variables in this study are less than 12%. Before running our analyses, we corrected for missing values in the data using multiple imputation (MI) chained equations in STATA by creating five imputed datasets and averaging the results across these data sets for final estimates (King, Honaker, Joseph, & Scheve, 2001).

In addition, the analyses were based on weighted samples that were created to adjust for oversampling bias and non-response (Ingels et al., 2011). Because the HSLs sampling design selected schools first and then students within these schools, we used the svy command in STATA with school ID as a cluster indicator and the transcript weight as a sampling weight. The base-year student weight (w1student) was used for ninth-grade achievement, and first-year student weight (w2w1stu) was used for eleventh-grade achievement. To comply with the NCES restricted use data requirements, we report unweighted sample sizes to the nearest 10 and report weighted means and proportions.

Measures

Dependent Variables The project has three important dependent variables—ninth- and eleventh-grade mathematics achievement scores, and course-taking. In particular, we used item response theory (IRT) scale test scores provided by the NCES to allow for the examination of growth over time (Ingels et al., 2011). IRT allows for cross-group comparison and was used to develop single proficiency scores across stages. We standardized these outcomes to a z score with a mean of 0 and pooled standard deviation (SD) of 1. The course-taking variable is a binary variable where 1 is where a student has taken Calculus or above and 0 where a student has not taken Calculus or above during high school. Since scholars construct the measure of course-taking differently based on their research, we also created variables such as Trigonometry or above, Algebra II or above, and Precalculus or above and ran the analyses. Though the effect sizes were slightly different, we found similar results and narratives to our original findings regardless of how we constructed the course-taking variable. However, in this study, we only report on Calculus-or-above results. In HSLs:09, about 3080 students were reported as having taken Calculus or above and about 14,670 as not taking Calculus or above during high school.

Early Algebra I To determine whether a student has taken Algebra I or above by eighth grade, we used students' survey responses and transcript information. The study found that 31% of the students had taken Algebra I or higher by the eighth grade, and 69% had taken it after the eighth grade. In addition, among the students who had taken algebra by the eighth grade, we have identified and accounted for students who repeated algebra in the ninth grade, which was about 3.4% of the students.

Covariates Student demographic information such as students' sex, race/ethnic group, English as a Second Language (ESL), and socioeconomic status (SES) was employed as student-level control variables. Students' math efficacy and math identity, their educational expectations and those of their parents, and parents' math efficacy were also taken into account. The study further includes additional variables, not only at the student level but also at the school level, to look more comprehensively at the relationship between early algebra and students' educational success. School-level variables include but are not limited to school size, school location, percentage of Free and Reduced Lunch, and school type. Lastly, we took account of students' final grade in their most advanced eighth grade math course. Variables in this study are listed in Table 1.

Analytic Plan

To examine the effect of early algebra on student achievement, we conducted three distinct analyses, naïve to more rigorous, to better understand and isolate the effectiveness of early algebra on student performance. For research question one, we used three forms of analysis: (1) OLS regression, (2) fixed effects models, and (3) propensity score matching (PSM), which will be discussed in detail below. For research question two, since the dependent variable for our second research question was binary—whether or not a student took Calculus or above during high school—we used the same approach, covariates, and the same considerations as research question one but ran it with a probit regression. Table 2 documents the mean and standard deviation (SD) of the variables in the study and presents the descriptive statistics between early algebra (by eighth grade) and algebra (post-eighth grade) students.

Model 1 - Regression (OLS and Probit) The following model (1) was utilized in this study for regression results:

$$Y_{ij} = \beta_0 + \beta_1 EA_{ij} + \beta_2 S_{ij} + \beta_3 SC_{ij} + \varepsilon_{ij} \quad (1)$$

where Y represents achievement outcomes (ninth- and eleventh-grade achievements) or course-taking (binary variable of whether a student took Calculus or not during high school)¹ by student i in school j . The model includes a primary dichotomous variable of early algebra (EA), a vector of student-level variables (S), and a set of school-level characteristics (SC). To account for the nesting structure of the data, we clustered the error at the school level.

Model 2 - State Fixed Effects Though we took a wide range of covariates into account in the first models, variables could have been omitted that could influence taking algebra by the eighth grade as well as students' educational outcomes. Some states may have fewer resources regarding mathematics curricula and therefore be less likely to offer algebra courses to their state students before high school. Thus, to hold unobserved time-invariant state-level factors constant, we employed state fixed effects. In other words, by including indicators at state level, this model held all state-by-state variation constant. To account for state indicators, the state fixed effects model includes δ_c in Eq. (1), which represents state fixed effects for students in state c in Eq. (2):

$$Y_{ij} = \beta_0 + \beta_1 EA_{ij} + \beta_2 S_{ij} + \beta_3 SC_{ij} + \varepsilon_{ij} + \delta_c \quad (2)$$

It is important to note that there were few states where all students had taken algebra before or after the eighth grade. These states were excluded from the analysis, giving an analytic sample of students for the state fixed effects model of about 15,350.

Model 3 - School Fixed Effects In addition, different quality, support, and resources at high school may influence students' ninth- and eleventh-grade achievement scores or

¹ For our probit regression, the following equation was estimated for student i in school j : $\Pr(\text{calculus or above}_{ij} = 1) = \Phi(\beta_0 + \beta_1 EA_{ij} + \beta_2 S_{ij} + \beta_3 SC_{ij} + \varepsilon_{ij})$

Table 1 Description of the variables in the study

Variables	Description
Ninth-grade achievement	IRT-based score on ninth-grade math test
Eleventh-grade achievement	IRT-based score on eleventh-grade math test
Course-taking	Dummy variable coded as 0 for students who have not taken Calculus or higher during high school and 1 for students who have taken Calculus or higher during high school
Sex	Respondent's gender (male or female). Dummy variable coded as 0 for female and 1 for male
Race/ethnicity	Categorical race/ethnicity variable, White being the reference category—Hispanic, African American, Asian, and Other variables were constructed.
ELL	Dummy variable coded as either 0 for non-ELL or 1 for ELL
SES	This composite variable is constructed to measure socioeconomic status. HSLS:09 used parent/guardians' education (X1PAR1EDU and X1PAR2EDU), occupation (X1PAR1OCC2 and X1PAR2OCC2), and family income (X1FAMINCOME) to construct the variable.
Math efficacy	This variable is a scale of the sample member's mathematics self-efficacy created by HSLS:09. Four questions have been used to composite this variable: "You are confident that you can do an excellent job on tests in this course (S1MTESTS)," "You are certain that you can understand the most difficult material presented in the textbook used in this course (S1MTEXTBOOK)," "You are certain that you can master the skills being taught in this course (S1MSKILLS)," and "You are confident that you can do an excellent job on assignments in this course (S1MASSEXCL)."
Math identity	This variable is a scale of the sample member's math identity created by HSLS:09. Students who tend to agree with the statements "You see yourself as a math person" and/or "Others see me as a math person" will have higher values in this variable.
Math GPA in the eighth grade	Final grade of most advanced eighth-grade math course.
Student expectation	Continuous measure of the student's expectation of education when in the ninth grade; range from 1 = less than high school graduation to 7 = obtain PhD, MD, or other advanced degree
Parental Expectation	Continuous measure of parents' expectation of the surveyed child's education when in the ninth grade; range from 1 = less than high school graduation to 7 = obtain PhD, MD, or other advanced degree.
Parent efficacy	Parents' confidence in helping students with math homework. 0 = Not at all confident, 1 = somewhat confident, and 2 = very confident
Public	Dummy variable coded as either 0 for private school or 1 for public school
School locale	Four dummy variables: city, suburban (reference), rural, and town
ELL percentage	Percentage of students who are ELL in a school
Special Ed. percentage	Percentage of students who are Special Ed. in a school
FRL percentage	Percentage of students who are FRL in a school
Minority percentage	Percentage of students of color in a school
AP percentage	Percentage of students enrolled in Advanced Placement courses in a school
Full-time math teachers	Number of full-time high school math teachers in a school

Table 2 Descriptive statistics of the variables in the study

Variables	Descriptive			
	Early Algebra ¹ mean (SD)		Algebra ² mean (SD)	
Ninth-grade achievement	47.4***	(11.7)	37.1	(10.6)
Eleventh-grade achievement	78.9***	(18.6)	62.7	(17.1)
Course-taking	0.4***	(0.5)	0.1	(0.2)
Sex	0.5*	(0.5)	0.5	(0.5)
Race/ethnicity				
White	0.6***	(0.5)	0.6	(0.5)
Black	0.1***	(0.2)	0.1	(0.3)
Hispanic	0.1***	(0.3)	0.2	(0.4)
Asian	0.1***	(0.3)	0.1	(0.2)
Others	0.1*	(0.3)	0.1	(0.3)
ELL	0.0***	(0.1)	0.0	(0.2)
SES	0.3***	(0.8)	− 0.1	(0.7)
Math efficacy	0.2***	(1.0)	0.0	(1.0)
Math identity	0.3***	(1.0)	− 0.1	(1.0)
Math GPA in the eighth grade	4.3***	(0.9)	3.9	(1.0)
Student expectation	8.3***	(2.4)	7.6	(3.1)
Parental expectation	7.8***	(2.1)	6.6	(2.4)
Parent's math efficacy	1.1***	(0.8)	1.0	(0.8)
Public	0.8***	(0.4)	0.8	(0.4)
School locale				
City	0.3***	(0.5)	0.3	(0.4)
Suburb	0.4**	(0.5)	0.4	(0.5)
Town	0.1***	(0.3)	0.1	(0.3)
Rural	0.2***	(0.4)	0.3	(0.4)
ELL percentage	4.9	(8.7)	4.7	(8.6)
Special Ed. percentage	10.7***	(7.3)	11.7	(7.8)
FRL percentage	28.4***	(23.4)	34.8	(25.1)
Minority Percentage	32.1**	(26.5)	33.5	(29.0)
AP percentage	19.3***	(15.5)	15.0	(12.3)
Full-time math teachers percentage	10.4***	(6.6)	9.9	(6.5)

Asterisks indicate statistically significant differences between students who enrolled in algebra by the eighth grade and students who enrolled in algebra during high school. Standard deviations are in parentheses

*** $p < .001$

** $p < .01$

* $p < .05$

¹ Early Algebra—students who enrolled in Algebra I by the eighth grade

² Algebra—Students who enrolled in Algebra I during high school (between the ninth and 12th grades)

*Unweighted descriptive statistics of the variables in the study

course-taking during high school. Similar to state fixed effects, the study additionally examined school fixed effects, where we are comparing students within a school and

which takes account of time-invariant variables at school level. Thus, we added δ_s which represents school fixed effects for students in schools. The following school fixed effects were employed to account for these biases in Eq. (3):

$$Y_{ij} = \beta_0 + \beta_1 EA_{ij} + \beta_2 S_{ij} + \varepsilon_{ij} + \delta_s \quad (3)$$

Similarly, as above, there were schools where all students in a school had taken algebra either by the eighth grade or during high school. These schools were excluded, which gave us an analytic sample of about 12,020.

Model 4 - Propensity Score-Matched Design Lastly, we used propensity score matching (PSM) developed by Rosenbaum and Rubin (1983) as an alternative for estimating the effects in the absence of randomized data. Previous literature used national representative dataset controls for many explanatory factors to isolate the relationship between the primary variable interest and their outcomes. In most cases, however, there is still a selection issue that cannot be removed by adding more covariates to the model. Thus, many scholars using non-experimental data use PSM to address selection bias on observable characteristics. For this reason, by identifying non-early algebra students who are most similar on observable characteristics and have taken algebra before high school on available observed characteristics, we created two groups with similarly observed covariates. Thus, as a supplementary analysis, the study investigates the effects of early algebra on student achievement using PSM.

The first step in PSM is to select the covariates upon which the matching will be based on (Augurzy & Schmidt, 2001). We selected theoretically relevant variables based on the literature that would conceivably predict taking algebra before high school. These variables have been identified in research as appearing to be significantly related to mathematics course-taking and achievement (Gamoran & Hannigan, 2000; Howard, Romero, Scott, & Saddler, 2015). These variables are the students' characteristics and their background information (e.g. gender, race, SES, English language learner), their academic traits (e.g. middle school math GPA, math efficacy, math identity), their family (e.g. parents' educational expectation, parents' math efficacy), and demographic and socioeconomic characteristics of the school districts in the Stanford Education Data Archive, which compiled the district data information from the Education Demographic and Geographic Estimates and the Common Core of Data. As Rubin and Thomas (1996) pointed out, accounting for potentially relevant covariates (as much as possible) will not hinder the quality of the PSM. The real threat is that by not including possible relevant covariates, the analysis will create bias.

Normally, studies using PSM use information prior to the treatment to estimate the propensity to be in algebra before high school. Though HSLS:09 provides and contains detailed information about the experiences of students and their secondary education, the variables were collected in the ninth grade, which is after the 8th-grade algebra treatment. As with Gottfried and Le's paper (2016), we made a similar assumption for this study that these measures—for instance, the students' math efficacy interest, their socioeconomic status, or the district's percentage of FRL—were fairly time-invariant. Empirically, we found that the measures remained relatively the same between the different data collection periods (ninth and eleventh grades). Nonetheless, we caution about any causal inferences suggested by the findings.

After missing imputation, we used the following logit model (4), based on Rosenbaum and Rubin (1983), to first estimate the propensity score²:

Log $[P(Z)/(1 - P(Z))] = \alpha + \beta f(Z)$ (4), where $\alpha + \beta$ were parameters and $f(Z)$ was the specified function. After estimating the propensity score, we used stratification strategy, which naturally incorporates survey weights from complex survey designs (Zanutto, Lu, & Hornik, 2005). According to Cochran (1968) and Imbens (2004), five strata are sufficient to remove about 90% of the bias. Using 32 strata, we were able to achieve a complete balance of all utilized independent variables in the matching procedure.³ Then, stratum, as a grouping factor, is added to the model to compare achievement between students who have taken algebra before and after the eighth-grade graduation for each propensity score stratum. As a result, the average treatment effect of early algebra could be estimated by re-analyzing the original regressions from model (1) but now by accounting for strata and controlling for propensity score (see e.g. Gottfried & Le, 2016; Hong & Raudenbush, 2005; Shifrer, Callahan, & Muller, 2013).

It is important to note that the propensity score matching used in this study accounts only for the selection bias in variables observed in the HSLs:09 dataset. Unobserved variables that affect both taking algebra early and the outcome challenge our ability to make causal inferences (Stuart & Rubin, 2008). In other words, the PSM model appears to reduce the bias on observed characteristics compared with traditional regression, but it is difficult to rule out the possibility that unobserved confounders may undermine our results (Alvarado & An, 2015). Nevertheless, by using the most recent nationally representative data, we exhausted the available resources to provide the most rigorous evidence for the effect of early algebra on the students' education.

Results

Overall, in all our analytic models, we found a positive significant relationship between early algebra and students' achievements and course-taking. This consistency suggests the findings of this study are robust. Since we standardized the coefficients, these effect sizes are comparable with each other and to other previous research. For purposes of economy, we have summarized only the effect sizes of early algebra on students' educational outcomes in Tables 3 and 4. Full results can be found in Appendix A and Appendix B as electronic supplementary material.

Association Between Early Algebra and Student Achievement

The findings show that students who have enrolled in algebra by the eighth grade had higher mathematics spring test scores in both ninth and eleventh grades, as compared with those who take algebra in high school. Specifically, in Table 3, accounting for a wide range of covariates, OLS results show that early algebra is positively and significantly ($p < 0.001$)

² Multiple imputation has been shown to adequately address the missing observation in the data when performing PSM (Mattei, 2009). We conducted matching and estimated our results independently on each of the five imputed datasets, and then pooled the results according to Rubin's rule (Rubin, 1987).

³ We ran the analysis with both five strata and 32 strata, and found similar trends and narratives. In this paper, we report results based on 32 strata.

associated with an increase in ninth-grade mathematics scores by $.59\sigma$ and an increase in eleventh-grade mathematics scores by $.56\sigma$. We found a similar narrative in the state and school fixed effects models, standard errors, and effect sizes. In the state fixed effects model, students who had taken algebra early showed significantly ($p < 0.001$) higher math achievement in the ninth ($.59\sigma$) and eleventh grades ($.57\sigma$), as compared with students who missed that experience. Likewise, in the school fixed effects model, the association between early algebra and student achievement was positively significant ($p < 0.001$) in both ninth-grade ($.64\sigma$) and eleventh-grade ($.62\sigma$) test scores. These results highlight that, after we accounted for unobserved time-invariant school- and state-level factors, the association between early algebra and achievement was still noticeable, and even stronger in the school fixed-effects model. Lastly, in our PSM model, we found that students' having taken early algebra was associated significantly ($p < 0.001$) with an increase of 0.58σ in ninth-grade and 0.55σ in eleventh-grade test scores. The magnitudes of the effect sizes as well as the standard errors were similar to OLS.

Association Between Early Algebra and Students' Course-Taking

We found that students who have taken algebra by the eighth grade are more likely to take advanced mathematics courses. To ease the interpretation, in Table 4, the probit regression results are reported as an average of the marginal effect of course-taking variables for a student with mean characteristics. We found that the predicted probability of taking Calculus or above is 17% higher for individuals who have taken algebra by the eighth grade compared with students who take algebra during high school. As exemplified in our models, this finding was similar across all our methodological approaches. Specifically, in state fixed effects models, taking algebra by the eighth grade was positively associated with an increased probability of advanced course-taking by 14% compared with students who take algebra after the eighth grade. Similarly, in a school fixed effects model, the likelihood of taking Calculus or above for early algebra students was 29% higher than for comparable students who took algebra during high school. In other words, within the same school, students who took algebra before high school had about a 30% higher likelihood of taking Calculus or above compared with students who took algebra in high school. Lastly, in our PSM model, we have found that students who have taken early algebra are 17% more likely to enroll in Calculus or above during high school compared with students who have taken algebra after the eighth grade.

Table 3 Effect sizes of early algebra on students' achievements

	Regression	State fixed	School fixed	PSM
Ninth-grade achievement	0.59*** (0.02)	0.59*** (0.02)	0.64*** (0.02)	0.58*** (0.02)
Eleventh-grade achievement	0.56*** (0.02)	0.57*** (0.02)	0.62*** (0.02)	0.55*** (0.02)

Clustered standard errors in parentheses. Full results can be found in Appendix A

* $p < 0.05$

** $p < 0.01$

*** $p < 0.001$

Table 4 Effect sizes of early algebra on students' course-taking

	Probit regression	State fixed	School fixed	PSM
Course-taking (Calculus) ^b	0.17***(0.01)	0.14**(0.00)	0.29***(0.01)	0.17***(0.01)

Coefficients are presented as probability. Clustered standard errors in parentheses. Full results can be found in Appendix B

* $p < 0.05$

** $p < 0.01$

*** $p < 0.001$

Discussion

Our empirical evidence documents algebra's significant role in the educational pathways of students. While much research has focused on the relationship between early algebra and student achievement, there is little rigorous empirical evidence showing an association between early algebra and advanced math course-taking. In all of our analytic models, students who take algebra before high school are more likely not only to have higher achievement scores but also to enroll in Calculus during high school. In fact, while students may have similar interest, efficacy, and ability in math, we found that those who take algebra by eighth grade are more likely to take advanced math courses during high school. While our results may indicate that students who take algebra early are very likely to have higher math achievements and take more advanced math courses in high school, due to the non-randomization of our data and selection bias, it needs to be cautious of interpreting these relationships as causal. Moreover, aligning with previous literature, findings from our study do not imply that *all* students should take or can benefit from taking algebra by eighth grade, as pushing students who are not academically prepared actually does more harm than good (Clotfelter et al., 2015; Liang et al., 2012), which will be discussed further below. The findings from our study have implications for research, policy, and practice.

First, our findings align with prior research regarding a positive association between early algebra and student achievement (Gamoran & Hannigan, 2000; Rickles, 2013; Spielhagen, 2006). Specifically, we find that students who take algebra by eighth grade, compared with those who do not, tend to have higher math test scores not only in the ninth but also in the eleventh grade. This result coincides with earlier studies that used national datasets such as NELS and LSAY and found a positive relationship between early algebra and later student achievement (Ma, 2000; Rickles, 2013). Our finding aligns with the *gatekeeper* notion of algebra that students taking algebra by eighth grade tend to have higher achievement scores in mathematics (Dougherty et al., 2017; Liang et al., 2012). Although the findings of this paper align with most prior studies using national datasets (Gamoran & Hannigan, 2000; Rickles, 2013), they disagree with two recent studies that looked at the efficacy and effectiveness of the universal or accelerated algebra policy (Clotfelter et al., 2015; Liang et al., 2012). One explanation for this is that, in reality, students are sorted into different mathematical tracks which, in turn, have a significant effect on their readiness to take algebra before high school (Loveless, 2009). Students who are placed in an upper track are probably ready to take algebra before high school, and students in remedial or general math classes may be

less prepared to take it before eighth grade. Certainly, having all students taking algebra before a specific grade level may do more harm than good. Clotfelter et al. (2015) maintain that introducing algebra in middle school, “absent any wholesale reform in prealgebra math courses, would exacerbate inequality and yield adverse effects” (p. 185).

In addition, our findings confirm that students who have taken algebra before high school are more likely than their peers to enroll in higher-level math courses in high school, which were largely descriptive and correlational in the literature (Paul, 2005; Spielhagen, 2006). This aligns with a limited prior study in which the researchers found that students who have not taken algebra before high school may have little chance of reaching advanced courses such as Trigonometry and Calculus (Adelman, 1999; Adelman, Daniel, & Berkovits, 2003; Schneider et al., 1998; Smith, 1996). As mentioned above, those students who take algebra in high school can either double up on their courses or take online classes if they want to take Calculus, Statistics, or Trigonometry. School districts in California and North Carolina, for instance, provide an option for students to take Algebra II and Geometry in the same school year and so accelerate their course-taking sequence (Fensterwald, 2015; Henry, Barrett, & Marder, 2016). Many school districts also offer students the opportunity to take one or more classes online. However, unlike courses in the humanities, which are largely undifferentiated by topic, doubling up is particularly challenging in mathematics since higher-level math courses usually require students to have mastered concepts and skills covered in previous classes (Garfunkel & Mumford, 2011). Also, understanding the foundations of mathematics is challenging, requiring relatively high motivation and self-discipline (Barbour, 2010; Rice, 2006). Not surprisingly, as shown in our descriptive statistics in Table 1, only about 10% of students who started algebra in high school completed Calculus before graduation compared with 40% of early algebra students. Therefore, to increase the probability of students who are taking algebra during high school enrolling in Calculus, leaders in education need to recognize that math is highly differentiated and sequentially organized, and so provide well-thought-out intentional course sequences that can segue into advanced courses (Schneider et al., 1998).

Although the present study shows that taking algebra early is a strong predictor of how far students will go in mathematics by the end of high school, the literature cautions that we need to recognize algebra’s unequal access across different racial-ethnic groups (Cortes & Goodman, 2014; Rubin & Noguera, 2004). Prior literature has documented that, compared with White students, students of color are significantly overrepresented in general and remedial classes and underrepresented in advanced-level math courses during high school (Stein et al., 2011). Moreover, several studies have found that students in remedial or general math courses are more likely to have underprepared or less-qualified teachers (Kalogrides, Loeb, & Bêteille, 2013) and follow an easier curriculum (Hallinan, 2006), which makes it difficult for them to take advanced courses later. For these reasons, algebra has been called a “new civil rights” issue to be addressed by states and school districts in terms of educational equity (Edley, 2017; Moses, 1993). To address the disparity, educational policymakers and leaders have implemented universal or accelerated algebra policies with the aim of having more students of color taking algebra before high school or by a specified grade level (Domina et al., 2015). Nonetheless, as mentioned above, this neglects the tracking issues that led to different degrees of readiness for students taking algebra. As a result, districts that implemented such policies have seen an increased failure rate in algebra followed by an increased rate of repeating the course (Clotfelter et al., 2015; Liang et al., 2012).

This study, along with some prior research, argues that any conversation about algebra should be about *how* to strengthen the mathematics curriculum so that all students are prepared for algebra rather than about *when* they should study it (Morgatto, 2008). Scholars have said that a more thoughtful reform of the math curriculum can promote student readiness for algebra by eighth-grade and better prepare them for advanced math courses in high school (Burris, Heubert, & Levin, 2006; Schoenfeld, 1995). Recent studies have found that students who were taught to develop their algebraic thinking during elementary school are more likely to increase their algebra readiness in middle school (Blanton, Stephens, Knuth, Gardiner, Isler, & Kim, 2015; Ketterlin-Geller & Chard, 2011). Recognizing the importance of algebra, the Common Core State Standards for Mathematics (CCSSM) was introduced in 2012 with the recommendation that algebra concepts should be integrated into seventh- and eighth-grade general math, which changed the discourse from when to take algebra to what needs to be covered in math in those grades (Kohli, 2016). While CCSSM started a dialog about a set of sharable academic expectations for all students across states (Mathews, 2016), by 2015, nearly half of the states backed out from the CCSSM (Gewertz, 2015). This led to a significant variation between school districts on when and what needs to be covered in eighth-grade math and in algebra. Some districts have started to address this issue by focusing on reforming and strengthening their middle school math curriculum instead of students completing algebra in certain grades (Sawchuk, 2018). Dougherty et al. (2017) find that middle school math acceleration programs significantly increase readiness for algebra and improves college readiness. Hence, leaders in education need to revisit and rethink the mathematics curriculum to see how it can be structured to enable all students to learn algebra in a timely manner.

Although the present study contributes to our understanding of the effect of early algebra on educational success by using three distinct analyses on a large, nationally representative, longitudinal dataset, students and teachers are not randomly distributed in schools and classrooms. Simply because students who take algebra early are likely to achieve better outcomes, readers are cautioned about interpreting the findings as having a strong causal inference. Future research needs to consider pathways that lead up to early algebra as well as course-taking patterns that are characterized by strong racial and ethnic disparities.

Lastly, while we used information from student transcripts, not all courses labeled “algebra” are the same. Specifically, what is being taught (Adelman, 1994) and from which textbook may not necessarily mirror the course title (Cogan, Schmidt, & Wiley, 2001). As a result, learning can vary and lead to different course opportunities in high school. Future research needs to consider such variations and investigate the heterogeneity effects within early algebra students on their educational pathways.

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